

Workshop on Essentials of Mortality Investigation of Asian Elephant

Illustrated Notes with Data Analysis & Charts

5th–6th June 2026

Dharamjaigarh & Raigarh Van Mandals · Bilaspur Forest Circle · Chhattisgarh

Organised by: Chhattisgarh Forest Department

Technical Support: Wildlife Institute of India, Dehradun

Laboratory Partner: ICAR-IVRI, Izatnagar · NDVSU, Jabalpur

1. Workshop Overview

1.1 Context & Participants

This two-day workshop was convened by the **Chhattisgarh Forest Department** in partnership with the **Wildlife Institute of India (WII), Dehradun** to strengthen technical capacity in elephant mortality investigation across the Bilaspur Forest Circle. It brought together **84 participants** from the Forest Department and Veterinary Service Department of Chhattisgarh State.

1.2 Objectives

- **Biology, Behaviour & Disease** — Comprehensive understanding of elephant biology, behaviour and major fatal diseases.
- **Necropsy, Biosafety & Sample Handling** — Hands-on competency in field necropsy under biosafety and biosecurity protocols.
- **Surveillance, Disposal & Remediation** — Operational frameworks for active/passive health surveillance, carcass disposal and site decontamination.

1.3 Expected Outcomes

- **Field Necropsy Skills** — Position and systematically eviscerate heavy elephant carcasses in remote settings.
- **Sample & Lab Protocols** — Tissue fixative formulation and secure packaging for histopathology, toxicology and microbiology.
- **Surveillance & Forensics** — Field-validated disposal protocols, standardised forensic reporting and proactive health monitoring.

2. Elephant Population Status — Chhattisgarh

2.1 Population Growth Trend (2001–2026)

The elephant population in Chhattisgarh has grown from a founding dispersal of **24 individuals** (2001) to an estimated **451 elephants** in 2026 — a compound annual growth rate (CAGR) of ~13% over 25 years. Bilaspur Circle hosts the largest concentration.

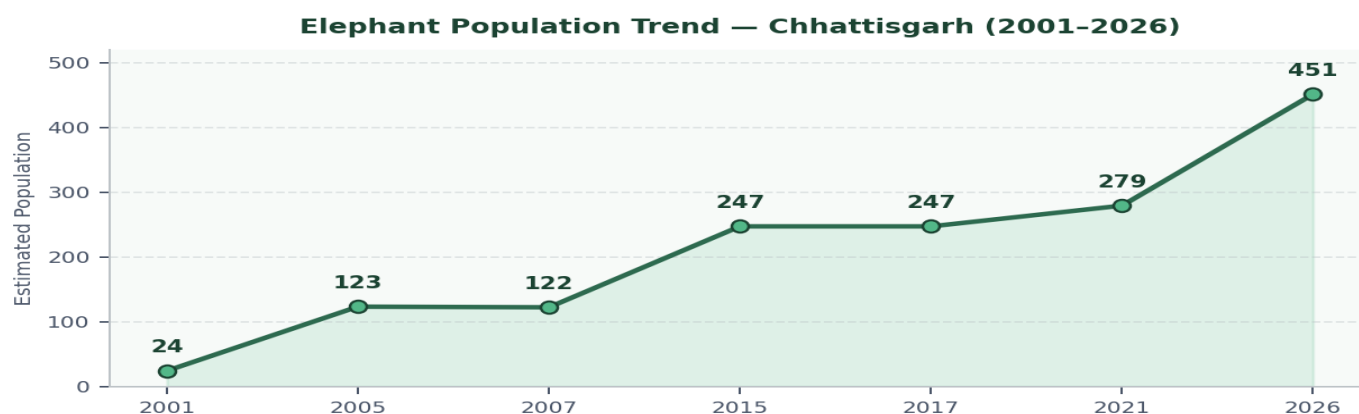


Figure 1: Elephant population trend 2001–2026 (Source: WII Census & CG Forest Dept. Estimates)

Year	Population	Notes
2001	24	Founding dispersal from Odisha/Jharkhand
2005	123	First major census count

2015	247	Significant increase
2021	279	Official Census (CAGR ~10% since 2005)
2026	~451 est.	Net +172 over 5-year study period

2025-26 records the highest single-year human-elephant conflict events in the five-year study period. Bilaspur Circle hosts the highest elephant density in CG.

3. Elephant Casualties — Five-Year Overview (2021–2026)

3.1 Year-wise Deaths — Division Comparison

49 confirmed elephant deaths across five study years plus ongoing 2026-27 data. Dharamjaigarh (DH) division accounts for 27 deaths; Raigarh (RG) division for 22.

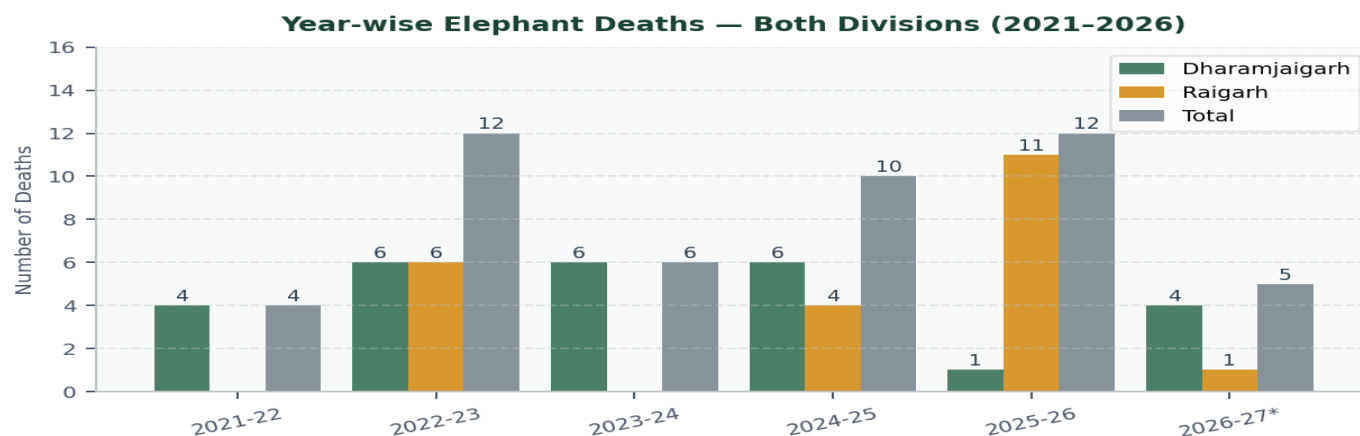


Figure 2: Year-wise elephant deaths by division. Bars show DH (green), RG (amber), Total (grey).

Year	DH Division	RG Division	Total	Notable
2021-22	4	0	4	All DH
2022-23	6	6	12	RG emerges; first cluster year
2023-24	6	0	6	DH dominant
2024-25	6	4	10	Both divisions active
2025-26	1	11	12	★ RECORD — RG peak
2026-27*	4	1	5	Ongoing — all calves, all drowning
TOTAL	27	22	49	5+ year study period

■ **2026-27 early signal: 5 deaths in first 3 months — all calves, all suspected drowning. Immediate field intervention required before peak dry-season period.**

3.2 Division-wise Elephant vs Human Casualties

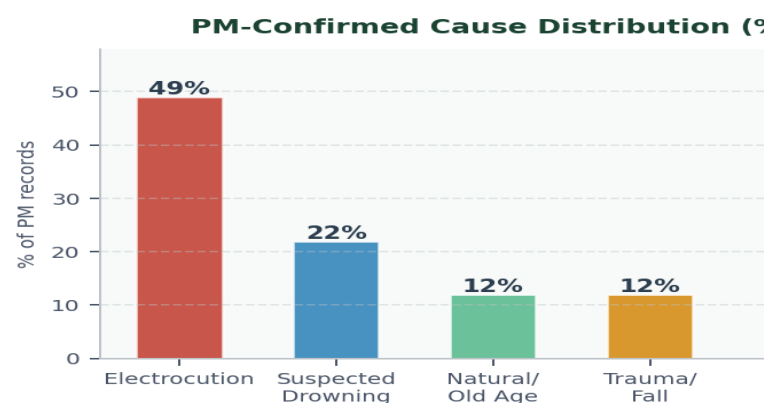
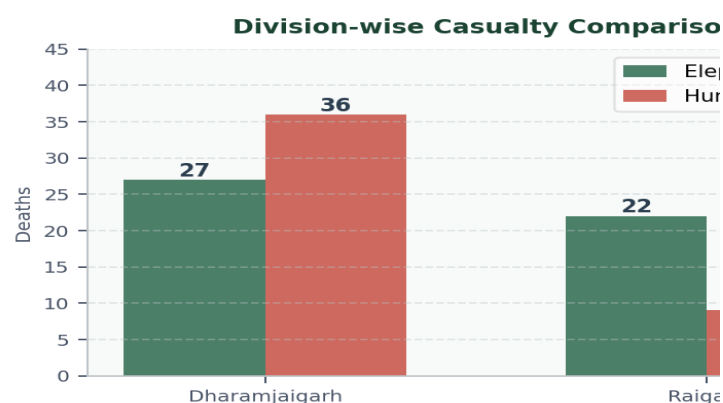


Figure 3 (left): Division-wise deaths. Figure 4 (right): PM-confirmed cause distribution.

Division	Elephant Deaths	Human Deaths
Dharamjaigarh	27	36
Raigarh	22	9
TOTAL	49	45

4. Cause of Death Analysis

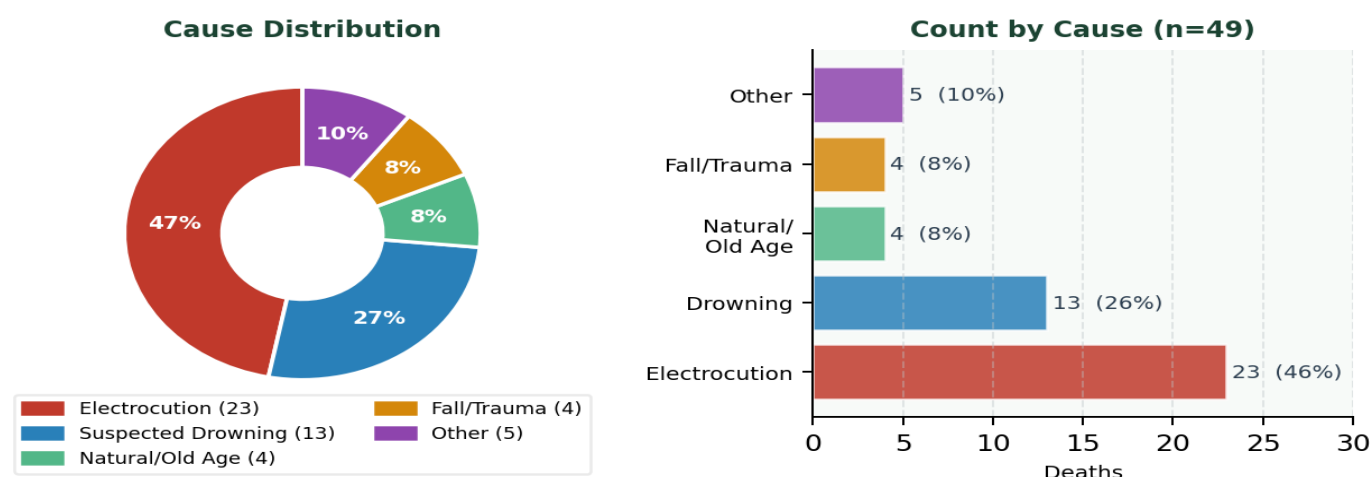


Figure 5: Cause of death — donut chart (left) and count with percentages (right). n = 49 deaths.

Cause of Death	Count	%	Key Risk Factor / Mechanism
Electrocution	23	47%	Illegal/low-hung power lines near crop fields; multi-simultaneous events in 3 cases
Suspected Drowning	13	27%	Dams, ponds, seasonal nalas; 92% are calves / young
Natural / Old Age	4	8%	Senescence; female-skewed (4F:1M)
Fall / Trauma	4	8%	Hills, nalas, inter-elephant infighting
Other (heat/sun)	5	10%	Heatstroke, weakness, undetermined

Electrocution (47%) + Suspected Drowning (27%) = 74% of all deaths. Both are preventable with targeted infrastructure and field intervention.

4.1 PM-Confirmed Cause Notes

- **20 PM records (49%)** confirm electrocution as cause of cardio-respiratory failure.
- Three cases show **multiple simultaneous electrocution events** — evidence of persistent illegal installations.
- **9 PM records (22%)** confirm suspected drowning; diatom test positive in all tested cases — confirmatory for ante-mortem drowning.

5. Demographic Analysis — Age & Sex Profile

5.1 Age Group Distribution

50% of all deaths are calves (0–2 yr) — the highest single age class. Combined with juveniles, **61% of deaths occur in individuals under 15 years**, posing a direct threat to long-term population viability.

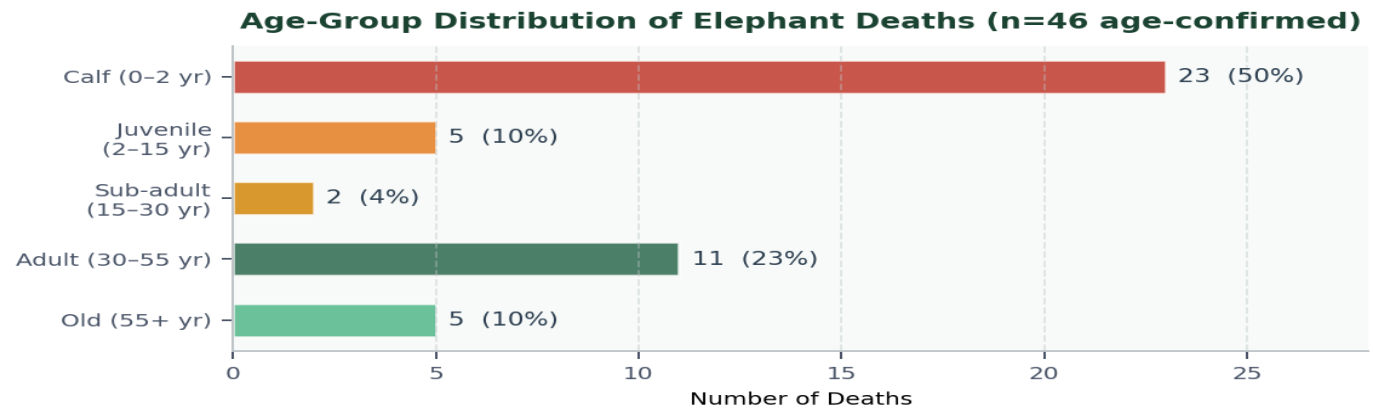


Figure 6: Age-group distribution of elephant deaths (n=46 age-confirmed). Calf class (red) accounts for half of all deaths.

Age Class	Count	%	Conservation Significance
Calf (0–2 yr)	23	50%	HIGHEST single class — critical population concern
Juvenile (2–15 yr)	5	11%	61% deaths under 15 yr combined
Sub-adult (15–30 yr)	2	4%	Low — transitional phase
Adult (30–55 yr)	11	24%	Electrocution primary; prime breeding cohort affected
Old (55+ yr)	5	11%	Female-skewed longevity

5.2 Sex Profile

Sex-wise Cause of Death Profile

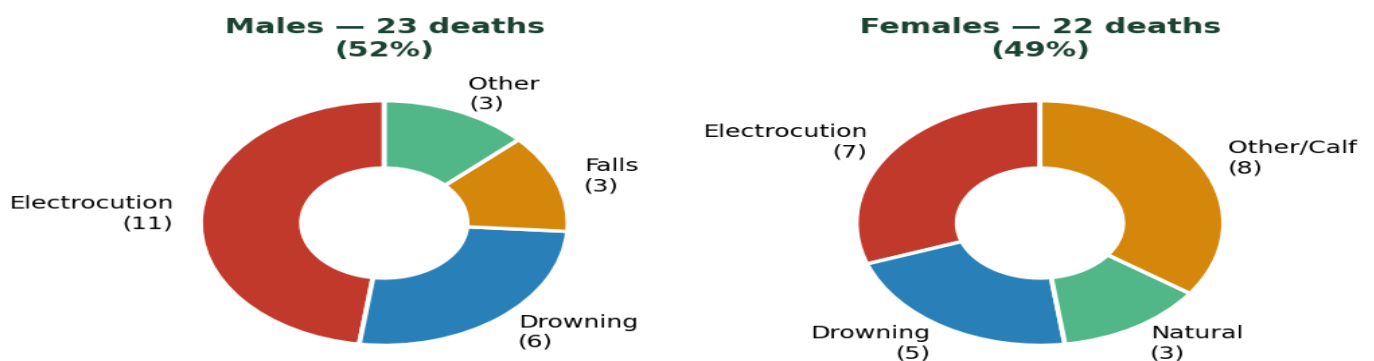


Figure 7: Sex-wise cause breakdown — males (left) and females (right). n = 45 sex-confirmed.

Sex	Count	%	Key Patterns
Male	23	52%	Wider age range (5–80 yr); electrocution highest (11 deaths)
Female	22	49%	Concentrated at extremes — 12 calves + 4 old; 55% of calf deaths female

- **Calf sex skew:** 55% of calf deaths are female. In Raigarh, 8 of 11 drowning deaths are young females (6 months–2 yr).

- **Male electrocution risk:** 48% of male deaths are electrocution vs 33% for females — males range farther into agricultural zones.
- **Female longevity:** More females survive to 55+ yr (4F vs 1M) — a natural pattern consistent with elephant biology.

6. Seasonal Variation — Monthly & Cause Patterns

6.1 Monthly Death Distribution

October is the deadliest month with 8 deaths, driven almost entirely by electrocution. The Oct–Dec window accounts for 42% of annual deaths. A secondary drowning peak occurs Jan–May during the dry season.

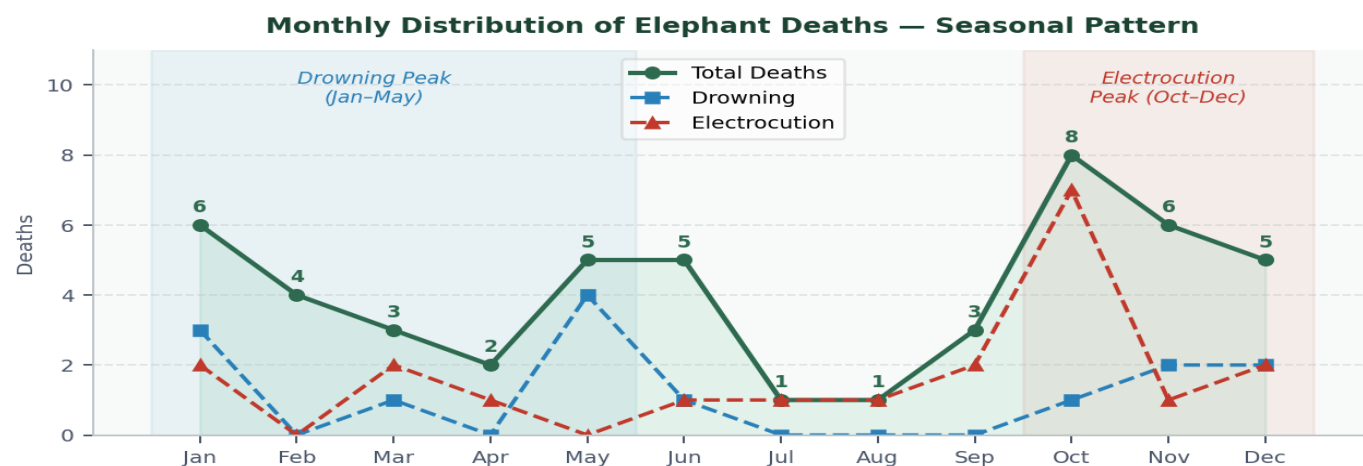


Figure 8: Monthly distribution of total deaths (green), drowning (blue dashed), electrocution (red dashed). Shaded zones mark seasonal peaks.

6.2 Season-wise Cause Breakdown

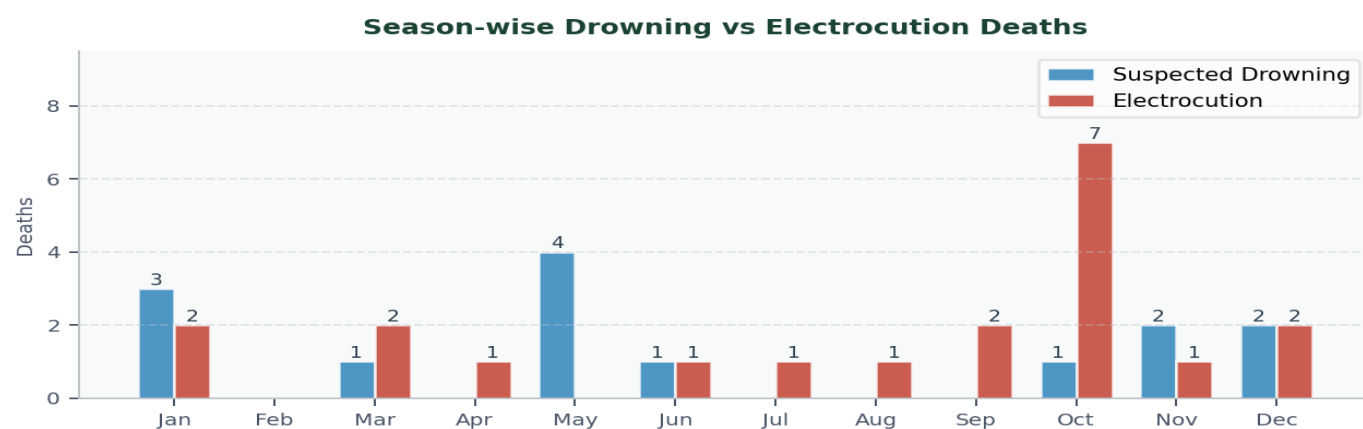


Figure 9: Monthly drowning vs electrocution deaths — highlighting opposing seasonal peaks.

Season	Drowning	Electrocution	Key Driver
Jan–May (Dry)	8 deaths (62%)	6 deaths (30%)	Low water, steep banks trap calves; crop fields quiet
Jun–Sep (Monsoon)	1 death (8%)	5 deaths (25%)	Nalas swell; some fencing active
Oct–Dec (Harvest)	4 deaths (31%)	10 deaths (50%)	Crop protection fencing at peak; post-monsoon movement

Drowning intervention window: Jan–May (62% of drowning deaths). Electrocutation intervention window: Oct–Dec (50% of electrocution deaths). Targeting these seasonal windows maximises intervention impact.

7. Range-wise & Geo Distribution

7.1 Range-wise Casualty Hotspots

Chhal (DH) and Ghargoda (RG) together account for **29 of 49 deaths (59%)**, making them **CRITICAL** intervention zones. Both show a mixed electrocution-drowning profile.

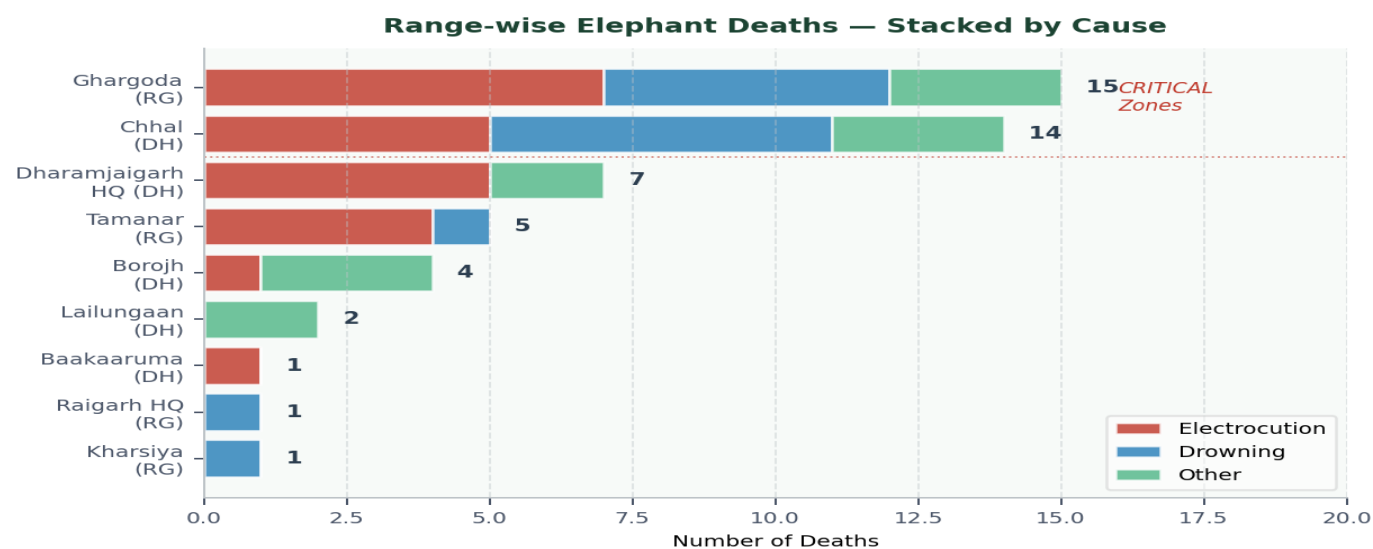


Figure 10: Range-wise deaths stacked by cause — electrocution (red), drowning (blue), other (green). Dashed line separates CRITICAL zones.

Range / Beat	Division	Total	Risk	Elec	Drown	Other
Ghargoda	Raigarh	15	CRITICAL	7	5	3
Chhal	Dharamjaigarh	14	CRITICAL	5	6	3
Dharamjaigarh HQ	Dharamjaigarh	7	HIGH	5	0	2
Tamanar	Raigarh	5	MODERATE	4	1	0
Borjoh	Dharamjaigarh	4	HIGH	1	0	3
Lailungaan	Dharamjaigarh	2	HIGH	0	0	2
Baakaaruma	Dharamjaigarh	1	MODERATE	1	0	0
Raigarh HQ + Kharsiya	Raigarh	2	LOW	0	2	0

7.2 Human Casualty Hotspots

Range / Beat	Division	Human Deaths
Chhal (DH)	Dharamjaigarh	10
Borjoh (DH)	Dharamjaigarh	9
Lailungaan (DH)	Dharamjaigarh	8
Ghargoda (RG)	Raigarh	5
Dharamjaigarh HQ	Dharamjaigarh	5
Others	Both	8

IMMEDIATE ACTION: Chhal & Ghargoda ranges — combined 29 elephant + 15 human deaths (59% of all elephant casualties). Multi-agency enforcement, power-line audit, and drowning infrastructure are required in these two ranges above all others.

8. Suspected Drowning — Detailed PM & Lab Analysis

13 deaths (27%) attributed to suspected drowning. **92% of victims are calves or young individuals.** The drowning cases cluster in Raigarh Division (dams, nalas) and Dharamjaigarh Division (ponds).

8.1 Gross Post-mortem Findings — All 13 Cases

PM Finding	Frequency / Notes
Frothy nasal/oral discharge — emphysema aquosum	ALL CASES — pathognomonic for drowning
Lungs — waterlogged, heavy, congested, oedematous	ALL CASES
Water in trachea & bronchi	ALL CASES
Gastric content — large volume of water & plant debris	ALL CASES
Skin maceration / hide wrinkling	CONSISTENT — prolonged water immersion
Petechial haemorrhages — conjunctiva / sub-pleural	VARIABLE

8.2 Histopathology — Microscopic Confirmation

- Pulmonary oedema — alveolar flooding with eosinophilic fluid.
- Pulmonary haemorrhage — interstitial and alveolar spaces.
- Hepatocyte degeneration — confirmed in **3 cases** (Tamnar Range, December 2025).
- **Diatom test POSITIVE** — confirmatory for ante-mortem (alive when entered water) drowning in all tested cases.
- Renal congestion — passive vascular engorgement.

■ **ARSENIC DETECTED — ICAR-IVRI Lab Report (File F.2-19/DI/NRC/2025-26/CWL) Arsenic at elevated levels in visceral samples from 2 specimens, Tamnar Range (Raigarh Division), December 2025. Histopathology confirmed hepatocyte degeneration. Arsenic likely a contributing metabolic stress factor from contaminated dam water near industrial corridor. Environmental sampling of Rabo / Panikshet dam catchments RECOMMENDED IMMEDIATELY.**

8.3 Toxicology — Negative Findings (5 tested batches)

Toxin Screened	Result
EEHV-I (Elephant Endotheliotropic Herpesvirus)	NEGATIVE — confirmed by ICAR-IVRI & NDVSU Jabalpur
Nitrate-Nitrite compounds	NEGATIVE
Organochlorine & Organophosphate insecticides	NEGATIVE
Heavy metals (Lead, Mercury, Cadmium)	NEGATIVE
Hydrogen Cyanide (HCN)	NEGATIVE

EEHV-negative result is particularly significant given the high neonate mortality. Viral haemorrhagic disease is ruled out as a contributing cause.

9. Workshop Programme & Resource Persons

9.1 Resource Persons

Name	Institution	Expertise
Dr. A. B. Shrivastav	NDVSU, Jabalpur	Wildlife Health, Forensics, Veterinary Pathology
Dr. Parag Nigam	WII, Dehradun	Wildlife Health, Chemical Capture, Conservation Medicine
Dr. Karikalan Mathesh	ICAR-IVRI, Bareilly	Wildlife Pathology, Histopathology, Molecular Diagnosis
Dr. Tapendra Saini	WII, Dehradun	Animal Genetics, Conservation Breeding, Wildlife Genetics
Dr. Chandra Prakash Sharma	WII, Dehradun	Wildlife Forensics, Morphological ID, Crime-Scene Management

9.2 Two-Day Programme at a Glance

Time	Session	Faculty
DAY 1 — 30 May 2026		
10:00–11:00	Inauguration	—
11:00–11:30	Elephant Status & Human-Elephant Interface in Chhattisgarh	PP, AM
11:30–12:15	Biological & Anatomical Aspects of Elephants	PN
12:15–13:00	Understanding Wildlife Mortalities: Veterinary Perspective	PN
14:00–15:30	Infectious Causes of Mortality (EEHV, FMD, TB, Anthrax, HS)	KM, ABS
15:30–16:30	Non-Infectious & Disaster Mortalities (Electrocution, Lightning, Train)	ABS, KM
16:30–17:30	Non-Infectious Diseases (Mycotoxycosis, Kodo Millet, Bio-toxins)	KM, PN
DAY 2 — 31 May 2026		
09:30–10:15	Biosafety & Biosecurity in Elephant Mortality	KM, TS, PN
10:15–11:00	Equipment Familiarization & Site Preparation	ABS, KM, TS
11:30–12:30	Comprehensive Sampling Protocols (tissue, cold chain, documentation)	KM, ABS, TS
12:30–13:00	Recording Information During Necropsy	KM, ABS
14:00–14:45	Vetro-Legal & Forensic Aspects	CP
14:45–16:00	Practical: Crime Scene Investigation & Ivory/Tusk Extraction	CP, PN, TS
16:15–16:45	Panel Discussion	All faculty
16:45–17:30	Valedictory	—

Faculty: ABS=Dr.Shrivastav · PN=Dr.Nigam · KM=Dr.Karikalan · TS=Dr.Saini · CP=Dr.C.P.Sharma · PP=Paras Patel · AM=Asst.Manager

10. Case Study — Gurda Calf Drowning · 01 June 2026

Date	01 June 2026
Location	Gurda, Kharsia Range, Raigarh Van Mandal
GPS	22.0439°N, 83.1313°E (22°3′16″N 83°8′11″E)
Elevation	256.04 ± 9.59 m
Water Body	Gurda River / exposed sandbank
Species	Asian Elephant (<i>Elephas maximus</i>)
Age Class	Calf (0–2 yr)
Sex	Not confirmed
Cause	Suspected Drowning

10.1 Field Observations

The calf was found on an exposed sandbank adjacent to the Gurda River in Kharsia Range. Photographic documentation was secured at 10:56 hrs (aerial view, 06-01-2026) and 11:03 hrs (ground-level carcass view). The incident is consistent with the established dry-season drowning pattern — when water levels are low and riverbanks are steep, young calves approaching the water edge to drink can fall in and are unable to climb out independently.

Original field note: ■■■■ ■■■■■■■■ ■■■■■■ ■■■■■■ — Gurda ■■■■■■■■■■, ■■■■■■ ■■■■■■■■ ■■■■■■■■

This case represents the 2026-27 cohort trend — all 5 deaths in the current year (first 3 months only) are calves, all suspected drowning. The Gurda case is the southernmost recorded drowning event in Raigarh Division (22.04°N).

11. Key Findings & Recommendations

11.1 Summary Statistics

Metric	Value
Study Period	2021-22 to 2026-27 (ongoing)
Total Elephant Deaths	49 confirmed
Leading Cause	Electrocution — 23 deaths (47%)
Second Cause	Suspected Drowning — 13 deaths (27%)
Most Vulnerable Age	Calves 0–2 yr — 23 deaths (50%)
Deadliest Year	2025-26 — 12 deaths (record)
2026-27 Early Signal	5 deaths in first 3 months — all calves, all drowning
Critical Hotspots	Chhal (14) + Ghargoda (15) = 59% of all elephant deaths
Drowning Seasonality	Jan–May = 62% of drowning deaths
Electrocution Seasonality	Oct–Dec = 50% of electrocution deaths
Arsenic Finding	Detected in 2 specimens — Tamnar Range, Dec 2025

EEHV Status	Negative — all batches tested
Toxicology	Negative — nitrates, organophosphates, heavy metals, HCN
Workshop Participants	84 (Forest Dept + Veterinary Service Dept, CG)

11.2 Priority Action Items

- **[IMMEDIATE] Environmental sampling** — Rabo & Panikshet dam catchments (arsenic detection follow-up).
- **[IMMEDIATE] Drowning infrastructure** — earthen ramps/exit points at high-risk dam edges in Chhal & Ghargoda.
- **[IMMEDIATE] Power-line audit** — survey of low-hung lines in all elephant corridors in critical ranges.
- **[Oct–Dec campaign]** — enforcement blitz on illegal crop-protection electric fencing during harvest season.
- **[Jan–May campaign]** — dry-season calf monitoring using GPS herd data; deploying watchers near known drowning sites.
- **Mandatory PM within 24 hrs** of mortality event + sample dispatch to ICAR-IVRI within 48 hrs with cold chain.
- Annual EEHV surveillance — document negative results systematically.
- Quarterly mortality review meetings at division level with all range officers.

Data Sources: Official Proforma-1 Records, Dharamjaigarh & Raigarh Divisions · PM Reports: ICAR-IVRI Izatnagar (File F.2-19/DI/NRC/2025-26/CWL) · GIS Data: Bilaspur Forest Division GIS Cell · Generated: June 2026

Organised by: Chhattisgarh Forest Department · Technical Support: WII, Dehradun · Laboratory: ICAR-IVRI & NDVSU, Jabalpur